Counterbleeding as Shared Activity Overapplication in Gua vowel harmony

Sören E. Tebay (Universität Leipzig)
tebay@uni-leipzig.de

Manchester Phonology Meeting 2025

Main Claims

- In Gua (Niger-Congo, Ghana) phrasal vowel harmony and phrasal vowel harmony interact in an opaque counterbleeding pattern (Obiri-Yeboah & Rasin 2024; OR24).
- I show that this pattern is expected if gradient symbolic representations with shared activity are assumed.
- This predicts that phrase-level opacity always involves spreading.

Phrase-Level Opacity

(1) Definition of Opacity (cf. Baković 2007)
A rule P applies opaquely if there are surface structures where
a. the P rule underapplies, e.g. there are contexts C for the rule to apply where the rule does not apply
b. or the rule P overapplies, e.g. there are structures derived by the rule P outside of its context C (≈counterbleeding).

(2) Counterbleeding
$P_{\text{triggers}} \leftarrow C_{\text{destroys}} \leftarrow Q$

- Phrase-level opacity poses a more severe challenge to parallel theories of phonology, such as Optimality Theory (Prince & Smolensky 1993)
- Even if extensions such as Strata (Kiparsky 2015, Bermudez-Otero 2012) or OO-Correspondence (Benua 1997) are added.

(3) **No Phrase-level Opacity Hypothesis**
A process X and a process Y cannot interact opaquely if both X and Y apply at the phrasal stratum.

Vowel Harmony and Hiatus resolution in Gua

- Gua cross-word ATR harmony applies in phonological phrases.
- A word-final [-ATR] vowel becomes [+ATR] if followed by a syllable with a [+ATR] vowel.

(4) Oral vowels (OR24:7)

+ATR		-ATR	
i	u	ɪ	ʊ
e	o	ɛ	ɔ
	a		ɜ

(5) Phrasal vowel harmony (OR24:8)
a. tʊ́ wátɕì — calabash break.PST — 'A calabash broke.'
b. t**ú** hè — calabash fall.PST — 'A calabash fell.'
c. kpʊ́tɔ̀ fíntɪ̀ — frog jump.PST — 'A frog jumped.'
d. kpʊ́t**ò** sʊ̀ — frog cry.PST — 'A frog cried.'

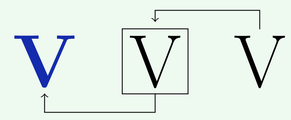
(6) Phrasal Hiatus resolution (OR24:12)
a. kwɛ̀lé téì — fry.IMP fish — 'Fry food!'
b. kwɛ̀l[ó] òní — fry.IMP meat — 'Fry fish!'
c. kpò òní — close fish
d. kp[ɛ̀] ésɛ̂ — close people

- Phrasal hiatus resolution includes total regressive assimilation of non-high vowels.

Opacity: vowel harmony and hiatus resolution

- The context for both processes can be present in an intraphrasal sequence of three words where the medial word is monosyllabic.
- The final vowel of the first word harmonizes with the **underlying** non-high vowel of the following word.

(7) Gua Counterbleeding



(8) Interaction between harmony and hiatus resolution (OR24:15,18)
a. /àŋɛ́ kwè ɛ̀dɛ̀/ → [àŋɪ**é** kw[ɛ̀] ɛ̀dɛ̀]
man grind.HAB something
'A man grinds something.'
b. /àhɛ́ tè ɔ̀kpʊ́kɔ́/ → [àh**é** t[ɔ̀] ɔ̀kpʊ́kɔ́]
knowledge slaughter.PST table
'Knowledge slaughtered a table.'

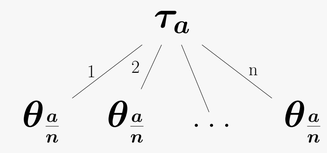
- Since interaction with other hiatus resolution strategies also involves vowel harmony, i.e. spreading, Shared Activity derives the patterns.

Gradient Analysis

- I propose that phrase-level opacity follows from Shared Activity (cf. Faust & Smolensky 2017, Zimmermann 2021) in gradient symbolic representations (Goldrick & Smolensky 2016).

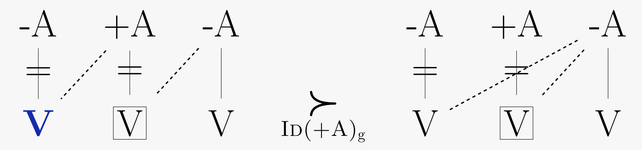
(9) **Shared Activity**
The activity a of a feature-bearing unit $\boldsymbol{\theta}$ depends on the number $\boldsymbol{n}$ of FBUs associated to the feature $\boldsymbol{\tau}$ that is associated to the FBU $\boldsymbol{\theta}$, i.e. $\boldsymbol{a(\theta) = \frac{a(\tau)}{n}}$.

(10) Feature spreading



- Processes are opaque if activity is changed by feature sharing.
- Resulting increase of faithfulness violations leads to overapplication.
- Spreading and coalescence involve sharing activity (cf. also Zaleska 2018) and thus increase the faithfulness violations.
- This means that spreading further (by coalescence) is worse than spreading less and shifting the [+ATR] feature for harmony.

(11) Effect of gradient faithfulness



IDENT(+ATR)$_g$: Count 1-$\boldsymbol{x}$ for a vowel with activity $\boldsymbol{x}$ that is associated to a [+ATR] feature in the input but not in the output.

Gradient Analysis: Vowel Harmony and Hiatus Resolution

- Vowel harmony follows from a directional markedness constraint *[-ATR]#[+ATR] and a split in faithfulness between IDENT(+ATR)$_g$ and IDENT(-ATR) (cf. OR24).
- A constraint against different cross-word mid vowels *[-h]$_i$#*[-h]$_j$ and a positional faithfulness constraint for initial vowels ID#V control vowel hiatus (cf. OR24).

(12) Phrasal Harmony as Shared Activity

Input=a.	*[-A]#[+A] 23	ID(+A)$_g$ 21	ID(-A) 1	$\mathcal{H}$
-ATR +ATR: a. tʊ́$_{1.0}$ hè$_{1.0}$	-1			-23
+ATR: ☞ b. t**ú**$_{.5}$ hè$_{.5}$			-1	-1

(13) Hiatus Resolution as Shared Activity

Input=a.	*[-h]$_i$#[-h]$_j$ 22	ID#V 22	ID(+A)$_g$ 21	ID(-A) 1	$\mathcal{H}$
+ATR -ATR: a. kpò$_{1.0}$ ɛ́$_{.5}$sɛ̃̂$_{.5}$	-1				-22
-ATR: ☞ b. kp[ɛ̀]$_{.\bar{3}}$ ɛ́$_{.\bar{3}}$sɛ̃̂$_{.\bar{3}}$			-0.$\bar{3}$		-7

Gradient Analysis: Opaque Interaction

- Unviolable OCP (Leben 1973, Goldsmith 1976, Myers 1997) bans adjacent identical autosegmental features.
- More sharing of activity in the transparent candidate in (14-c).
- More violation of the gradient faithfulness constraint IDENT(+ATR)$_g$.

(14) Counterbleeding as Gradient Faithfulness and Shared Activity

Input=a.	*[-A]#[+A] 23	*[-h]$_i$#[-h]$_j$ 22	ID(+A)$_g$ 21	ID(-A) 1	$\mathcal{H}$
-ATR +ATR -ATR: a. à$_{.5}$hɛ́$_{.5}$ tè$_{1.0}$ɔ̀$_{.\bar{3}}$kpʊ́$_{.\bar{3}}$kɔ́$_{.\bar{3}}$	-1	-1			-45.00
-ATR +ATR -ATR: b. à$_{1.0}$h**é**$_{.5}$ tè$_{.5}$ɔ̀$_{.\bar{3}}$kpʊ́$_{.\bar{3}}$kɔ́$_{.\bar{3}}$		-1		-1	-23.00
-ATR: c. à$_{.1\bar{6}}$hɛ́$_{.1\bar{6}}$t[ɔ̀]$_{.1\bar{6}}$ɔ̀$_{.1\bar{6}}$kpʊ́$_{.1\bar{6}}$kɔ́$_{.1\bar{6}}$			-0.8$\bar{3}$		-17.50
-ATR +ATR -ATR: ☞ d. à$_{1.0}$h**é**$_{1.0}$t[ɔ̀]$_{.25}$ɔ̀$_{.25}$kpíʊ$_{.25}$kɔ́$_{.25}$			-0.75	-1	-16.75

REFERENCES: **Baković**, Eric. 2007. A revised typology of opaque generalisations. Phonology 24(2). 217–259. **Benua**, Laura. 1997. Transderivational identity: Phonological relations between words. Amherst, MA: University of Massachusetts dissertation. **Bermúdez-Otero**, Ricardo. 2012. The architecture of grammar and the division of labour in exponence. In Jochen Trommer (ed.), The morphology and phonology of exponence, 8–83. Oxford: Oxford University Press. **Faust**, Noam & Paul **Smolensky**. 2017. Activity as an alternative to autosegmental association. Talk given at Manchester Phonology Meeting 25. **Goldrick**, Matthew & Paul **Smolensky**. 2016. Gradient symbolic representations in grammar: The case of French liaison. Available online at Rutgers Optimality Archive: roa.rutgers.edu/article/view/1552. http://roa.rutgers.edu/. **Goldsmith**, John. 1976. Autosegmental phonology. Cambridge, MA: MIT dissertation. **Kiparsky**, Paul. 2015. Stratal OT: A synopsis and FAQs. In Yuchau E. Hsiao & Lian-Hee Wee (eds.), Capturing phonological shades within and across languages, 1–45. Newcastle: Cambridge Scholars Publishing. **Leben**, William. 1973. Suprasegmental phonology. Cambridge, MA: Massachusetts Institute of Technology dissertation. **Myers**, Scott. 1997. OCP effects in Optimality Theory. Natural Language & Linguistic Theory 15(4). 847–892. **Obiri-Yeboah**, Michael & Ezer **Rasin**. 2024. Productive phrasal opacity in Gua: A challenge to stratal optimality theory. Natural Language & Linguistic Theory. **Prince**, Alan & Paul **Smolensky**. 1993. Optimality Theory: Constraint interaction in generative grammar. Available online at Rutgers Optimality Archive: roa.rutgers.edu/files/537-0802/537-0802-PRINCE-0-0.PDF. **Zaleska**, Joanna. 2018. Coalescence without coalescence: Universität Leipzig dissertation. **Zimmermann**, Eva. 2021. Faded copies: Reduplication as distribution of activity. Glossa: a journal of general linguistics 6(1).